

Functional Requirements Document (FRD)

TeamMatch

SkillSwap Team

1. System Overview

TeamMatch is a deterministic, constraint-driven team formation system designed for project-based computer science courses. The system operationalizes research-backed principles of heterogeneous grouping and balanced skill distribution to improve fairness, collaboration quality, and learning outcomes.

The system supports:

- Instructor-driven team constraints
 - Structured student skill modeling
 - Deterministic team assignment
 - Transparent explanation generation
-

2. System Context

Primary Users

- Computer Science Instructors (Doctor/Professor/Adjunct)

Secondary Users

- Undergraduate/Community College Students in project-based courses

System Role

TeamMatch acts as a decision-support tool for instructors. It does not replace instructor authority but generates optimized team assignments based on structured input and defined constraints.

3. Functional Requirements

3.1 User Roles & Access Control

FR-1: The system shall support two user roles:

- Instructor
- Student

FR-2: The system shall restrict team formation configuration functionality to Instructor users only.

FR-3: Students shall only be able to:

- Submit input data (skills, experience, availability, role preference)
 - View final team assignment (if released by instructor)
-

3.2 Data Collection

Student Input

FR-4: The system shall collect structured student information including:

- Self-reported technical skills
- Experience level per skill (e.g., beginner/intermediate/advanced)
- Weekly availability schedule
- Preferred project role (optional)

FR-5: The system shall validate student inputs for completeness before submission.

FR-6: The system shall allow instructors to define submission deadlines.

Instructor Input

FR-7: The system shall allow instructors to define:

- Desired team size
- Total enrollment
- Required or preferred skill categories
- Project-specific skill weighting
- Hard constraints (e.g., no solo teams, minimum 1 advanced per team)

FR-8: The system shall prevent invalid configurations (e.g., team size not dividing enrollment unless instructor allows remainder handling).

3.3 Skill Normalization

FR-9: The system shall convert unstructured skill descriptions into predefined standardized technical categories.

FR-10: The system shall map synonymous skills (e.g., “JS” → “JavaScript”) into a normalized skill taxonomy.

FR-11: The system shall assign numeric weight values to skills based on instructor-defined project emphasis.

3.4 Team Formation Engine

Deterministic Behavior

FR-12: The system shall produce identical team outputs for identical inputs and constraints.

FR-13: The system shall not use non-seeded randomness in team generation.

Constraint Enforcement

FR-14: The system shall enforce:

- Team size constraints
- Enrollment constraints
- Instructor-defined minimum/maximum skill thresholds
- Experience-level distribution constraints

FR-15: The system shall not generate any team that violates defined hard constraints.

Optimization Objectives

FR-16: The system shall minimize skill variance across teams.

FR-17: The system shall balance experience levels across teams.

FR-18: The system shall maximize availability overlap within each team.

FR-19: The system shall distribute high-experience students across teams rather than clustering them.

Assignment Logic

FR-20: The system shall implement a rule-based greedy assignment algorithm.

FR-21: The algorithm shall:

1. Prioritize placement of highest-impact students (e.g., advanced skill holders)
 2. Distribute scarce skills evenly
 3. Fill remaining slots to balance overall team scores
-

3.5 Team Evaluation Metrics

FR-22: The system shall compute for each team:

- Aggregate skill score
- Skill coverage index
- Experience balance metric
- Availability overlap score

FR-23: The system shall compute cross-team variance metrics.

FR-24: The system shall present comparison metrics against a random baseline grouping model.

3.6 Explanation & Reporting

FR-25: The system shall generate an instructor-facing report explaining:

- Why each student was assigned to a given team
- Which constraints influenced assignment
- Skill balancing justification

FR-26: The system shall generate visual or tabular summaries including:

- Team composition
- Skill distribution charts
- Experience distribution breakdown

FR-27: The system shall allow instructors to export reports in PDF or CSV format.

3.7 Reproducibility & Transparency

FR-28: The system shall log:

- Input datasets
- Instructor constraints
- Algorithm version
- Timestamp of generation

FR-29: The system shall allow instructors to regenerate teams using stored configuration.

FR-30: The system shall maintain version control over team formation logic.

3.8 Instructor Override

FR-31: The system shall allow instructors to manually adjust teams after generation.

FR-32: The system shall update evaluation metrics dynamically after manual changes.

FR-33: The system shall clearly indicate which assignments were system-generated versus manually modified.

4. Non-Functional Requirements

4.1 Determinism

- The system must guarantee reproducibility for auditing and fairness validation.

4.2 Performance

- Team formation shall complete within:
 - ≤ 5 seconds for classes up to 200 students.

4.3 Usability

- Instructors shall be able to configure and generate teams within 10 minutes.
- Interface must clearly distinguish hard vs soft constraints.

4.4 Explainability

- All algorithmic decisions must be traceable.
- No black-box optimization models (e.g., opaque ML without explanation).

4.5 Security & Privacy

- Student data shall not be publicly accessible.
 - Only instructors may view full class skill data.
-

5. Data Requirements

5.1 Student Entity

- StudentID
- Name
- SkillList[]
- SkillLevel[]
- AvailabilityMatrix
- PreferredRole

5.2 Instructor Configuration Entity

- CourseID
- TeamSize

- SkillWeights[]
- HardConstraints[]
- OptimizationPriorities[]

5.3 Team Entity

- TeamID
- MemberList[]
- SkillScore
- ExperienceScore
- AvailabilityScore
- VarianceMetrics

6. Traceability Matrix (PRD → FRD Alignment)

PRD Goal	FRD Reference
Heterogeneous skill distribution	FR-16, FR-19
Reduce instructor effort	FR-7, FR-20
Deterministic outputs	FR-12
Explainable system	FR-25
Skill normalization	FR-9
Reproducibility	FR-28
Research alignment	FR-16–FR-19

7. Assumptions

- Students provide truthful self-reported data.
 - Instructor-defined constraints reflect course goals.
 - Skill categories are predefined and maintained.
-

8. Future Extensions (Non-Core)

- Machine learning-based role prediction
 - Peer feedback integration
 - Longitudinal performance tracking
 - LMS integration (Canvas, Moodle, etc.)
-

End of FRD